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Risk-Adapted Starting Ages of Colorectal Cancer Screening for People With Diabetes or Metabolic Syndrome

Teresa Seum[1,2] | Michael Hoffmeister[1] | Hermann Brenner[1,3]

[1]Division of Clinical Epidemiology and Aging Research, German Cancer Research Center (DKFZ), Heidelberg, Germany | [2]Medical Faculty Heidelberg, Heidelberg University, Heidelberg, Germany | [3]German Cancer Consortium (DKTK), German Cancer Research Center (DKFZ), Heidelberg, Germany

Correspondence: Hermann Brenner (h.brenner@dkfz-heidelberg.de)

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ABSTRACT

Background: Individuals with diabetes and metabolic syndrome have an increased risk of colorectal cancer (CRC), suggesting earlier screening than the average-risk population may be warranted.

Aims: To derive risk-adapted starting ages of CRC screening for people with diabetes or metabolic syndrome.

Methods: We determined 5-year cumulative risks of CRC at individual ages between 30 and 50 across Europe (overall and individually for Germany, France, the UK and Italy) and North America (the United States and Canada) based on the GLOBOCAN 2022 database. Using risk estimates from meta-analyses (2020–2023), we derived the ages at which individuals with diabetes or metabolic syndrome reach the same CRC risk as the average-risk population at age 50 (aCR_{50}) or 45 (aCR_{45}).

Results: Individuals with diabetes were estimated to reach aCR_{50} at age 47 (95% confidence interval, 45–49) in Europe and 46 (42–49) in North America. For metabolic syndrome, the corresponding ages were 47 (47–48) in Europe and 46 (46–47) in North America. Disparities across countries were minimal, with deviations of up to no more than one year. For screening programmes starting at age 45, corresponding risk-adapted starting ages for people with diabetes or metabolic syndrome were estimated to be 42 (41–44) and 43 (42–43) for Europe and 41 (38–44) and 41 (41–42) for North America, respectively.

Conclusions: People with diabetes or metabolic syndrome reach risk levels comparable to the average risk population three to four years earlier. Our results offer empirical guidance for defining risk-adapted starting ages of CRC screening for these high-risk groups.

1 | Introduction

Colorectal cancer (CRC) is the second leading cause of cancer-related deaths worldwide [1]. The disease typically has a long development and preclinical phase, providing a crucial window for prevention and early detection through screening [2, 3]. Evidence from observational studies, randomised trials and modelling studies have shown that CRC screening can substantially reduce CRC incidence and mortality [4]. Consequently, screening programmes have been widely adopted in numerous countries [5]. In Europe, the most common age to begin screening is 50 years, though there is considerable variation, with some countries starting as early as 40 and others as late as 60 years [6]. In the US, recent epidemiological changes have led to the lowering of the recommended starting age for CRC screening to 45 years [7]. Furthermore, there is growing evidence of substantial variability in CRC risk within populations, which has led to calls for personalised, risk-adapted screening strategies

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[8]. This shift acknowledges the limitations of a one-size-fits-all approach and highlights the need for screening protocols that are more closely tailored to individual risk profiles. Established guidelines currently recommend earlier screening for individuals with a family history of CRC [9, 10]. However, while only about 25% of CRC cases are linked to family history, other significant risk factors are not routinely considered in screening recommendations [11].

Among these, diabetes and metabolic syndrome are recognised as significant CRC risk factors [12, 13]. These conditions are marked by insulin resistance, chronic inflammation and dyslipidaemia, which are associated with an increased incidence of CRC, including early-onset CRC [13, 14]. The exact mechanisms underlying this elevated risk are not fully understood but have been hypothesised to involve the effects of chronic inflammation, disruptions in the gut microbiota and the activity of insulin-like growth factors that support the proliferation of cancer cells [15, 16]. Furthermore, diabetes and metabolic syndrome are frequently accompanied by obesity and a sedentary lifestyle, which can exacerbate inflammation and stimulate the production of adipokines—factors that may facilitate tumour growth [17]. Recognising the heightened risk, individuals with diabetes or metabolic syndrome could benefit from starting CRC screening at a younger age. However, translating this increased risk into practical screening recommendations remains a major challenge. In this study, we illustrate how data on elevated risk can be used to determine risk-adapted starting ages for CRC screening in a straightforward way, with the aim of enhancing early detection and improving outcomes for those with metabolic disorders.

2 | Materials and Methods

2.1 | Data Sources

To assess the CRC risk associated with diabetes and metabolic syndrome, we extracted data from the latest meta-analyses which we identified by a systematic literature search as outlined in the Supporting Information. Whenever available, we specifically sought out meta-analyses that differentiated between different types of diabetes and definitions of metabolic syndrome to ensure a more detailed understanding of the associated risks. We furthermore specifically searched for meta-analyses focusing on the associations of diabetes and metabolic syndrome with early-onset CRC. The selected meta-analyses were then critically appraised for their relevance and comprehensiveness, focusing on the most recent studies and those including the most relevant data.

We extracted the latest and most comprehensive data on the absolute risk of CRC (ICD-10 codes C18-C21) by age in Europe (overall and specifically for the four European countries with the largest population: Germany, France, the United Kingdom (the UK), and Italy) and North America (overall and separately for the United States of America (the US) and Canada) from the GLOBOCAN 2022 database (https://gco.iarc.fr/today/en) [18]. Data sources of the GLOBOCAN database for the populations included in this analysis are described in Supporting Information Table S1. We extracted cumulative risks of CRC within 5-year age intervals 30–34, 35–39, 40–44, 45–49 and 50–54 which quantify the risk of a CRC diagnosis over the next five years at ages 30, 35, 40, 45 and 50, respectively.

2.2 | Analyses

Our analyses incorporated the average cumulative risk of CRC within the next 5 years at ages 30, 35, 40, 45 and 50, across Europe and North America, and in the most populous countries within these regions. Risks for those with diabetes or metabolic syndrome were calculated by multiplying average risks with the pooled relative risk estimates from the selected meta-analyses. We calculated 5-year cumulative risks at single years of age from age 30 to 50 by linear interpolation between cumulative risks at ages 30, 35, 40, 45 and 50, and we generated graphical plots of the 5-year cumulative risks for each age between 30 and 50 for the average risk population and people with diabetes or metabolic syndrome. In these plots, we added horizontal lines at the 5-year cumulative risk levels at age 50 (aCR_{50}) or age 45 (aCR_{45}) (the recommended starting age of CRC screening in the US) of the average risk population. We determined risk-adapted starting ages for people with diabetes or metabolic syndrome as the ages at which their cumulative risk curves intersected with these horizontal lines. We calculated 95% confidence intervals (CIs) for these ages by replicating our analyses with the upper and lower ends of the 95% CIs of the relative risk estimates from the meta-analyses. All data analyses and data visualisations in this study were done using Python 3.11.7.

3 | Results

An overview of the meta-analyses quantitatively assessing CRC risk associated with diabetes and metabolic syndrome is given in Table [1]. Using data from 47 studies, including 104,351 CRC cases and 18,416,872 controls, the pooled relative risk (RR) for type 2 diabetes and CRC was found to be 1.29 (95% CI, 1.23–1.36) in a meta-analysis from 2020 by Ling et al. [19]. The association between metabolic syndrome and CRC was assessed in a meta-analysis by Shen et al. [20]. Shen et al. included studies in which metabolic syndrome was defined as the presence of three or more metabolic abnormalities. These included increased waist circumference, elevated blood pressure, high fasting triglyceride levels, low HDL cholesterol levels and elevated fasting blood sugar [20, 21]. Based on 17 studies, a pooled RR of 1.36 (95% CI, 1.26–1.47) was reported. For the outcome of early-onset CRC, Khoa Ta et al. [22] synthesised data from 19 studies in 2023, identifying a pooled odds ratio (OR) of 1.43 (95% CI, 1.08–1.89) for those with diabetes. At the time of manuscript preparation, there was a lack of sufficient numbers of studies and cases to assess the risk for individuals with metabolic syndrome for early-onset CRC. Notably, high levels of heterogeneity, as reflected in the large I^2 values, were observed and taken into account by random-effects estimates in all meta-analyses.

The average 5-year cumulative risks for CRC at ages 30, 35, 40, 45 and 50 for all of Europe and specifically for Germany, the UK, Italy, and France, as well as North America, with separate data for the US and Canada are presented in Table 2. In both Europe and North America, 5-year cumulative CRC risks

TABLE 1 | Overview of the results of the selected meta-analyses.

Ref.	Predictor	Outcome	Numbers of participants		Results		
			(eo) CRC cases	Controls	Studies (datasets)	Risk estimate Pooled RR/OR (95% CI)	Heterogeneity I^2 (%)
Ling et al., 2020 [19]	T2D	CRC	104,351	18,416,872	47	1.29 (1.23, 1.36)	88.6
Shen et al., 2021 [20]	MetS	CRC	199,605	29,809,033	17 (27)	1.36 (1.26, 1.47)	86.1
Khoa Ta et al., 2023 [22]	DM	eoCRC	33,359	14,259,289	12	1.43 (1.08, 1.89)	95.1

Abbreviations: CI, confidence interval; CRC, colorectal cancer; DM, diabetes; eoCRC, early-onset colorectal cancer; MetS, metabolic syndrome; OR, odds ratio; RR, risk ratio; T2D, type 2 diabetes.

TABLE 2 | 5-year cumulative risk of colorectal cancer in Europe and North America in 2022 according to age.

		5-year cumulative risk from age on [%]							
	Age	Europe	Germany	France	United Kingdom	Italy	North America	United States	Canada
Total	30	0.01	0.01	0.02	0.02	0.01	0.04	0.04	0.03
	35	0.02	0.01	0.04	0.03	0.03	0.07	0.08	0.06
	40	0.06	0.05	0.09	0.07	0.07	0.14	0.14	0.11
	45	0.13	0.10	0.16	0.13	0.13	0.22	0.22	0.18
	50	0.24	0.19	0.26	0.24	0.22	0.33	0.34	0.27

(*Source:* GLOBOCAN 2022).

strongly increased with age, from 0.01% to 0.24% and from 0.04% to 0.33%, respectively. All countries exhibited a strong age-related risk increase, but the rate of increase and the absolute risk values at age 50 varied somewhat. The steepest increase and highest 5-year risk at age 50, as well as at younger ages, were observed in the US.

Figure 1 illustrates the estimated cumulative 5-year risks of CRC based on relative risk estimates at any age for individuals with diabetes and metabolic syndrome at ages 30, 35, 40, 45 and 50, depicted by coloured dots. Additionally, the figure shows the interpolated cumulative 5-year risks at the intermediate ages, represented by coloured lines, for both Europe and North America. The horizontal red line marks the cumulative 5-year CRC risk at age 50 (aCR_{50}) for the average risk population, while the cumulative 5-year CRC risk at age 45 (aCR_{45}) for the average risk population is indicated by the purple line. Considering relative risk estimates for CRC at any age, those with diabetes reached the aCR_{50} (0.24% for Europe, 0.33% for North America) at age 48 (95% CI, 47–48) in Europe and at age 47 (95% CI, 46–47) in North America. For people with metabolic syndrome, aCR_{50} was reached at age 47 (95% CI, 47–48) in Europe and at age 46 (95% CI, 46–47) in North America. With regard to screening programmes that start at age 45, people with diabetes reached aCR_{45} at age 43 (95% CI, 43–43) in Europe and at age 42 (95% CI, 41–42) in North America. For those with metabolic syndrome, aCR_{45} was reached at age 43 (95% CI, 42–43) in Europe and age 41 (95% CI, 41–42) in North America. Country-specific risk curves are detailed for Germany, France, the UK, Italy, the US and Canada in Supporting Information Figure S1.

Figure 2 shows analogous risk curves for the average risk population in Europe and North America and people with diabetes in which the latter were derived based on specific risk estimates for early-onset CRC. According to this analysis, individuals with diabetes reached aCR_{50} at age 47 (95%, 45–49) in Europe and at age 46 (95%, 42–49) in North America. Further, they reached aCR_{45} at age 42 (95% CI, 41–44) in Europe, and at 41 (95% CI, 38–44) in North America. Country-specific risk curves are shown in Supporting Information Figure S2.

The risk-adapted starting ages for screening, calculated using the average 5-year risk at the benchmark initial screening age of 50 as well as at age 45 for each country are summarised in Table 3. Although age-specific incidence rates varied, the risk-adapted starting ages were fairly consistent across the assessed populations. The aCR_{50} was quite consistently reached at ages 46 to 48 years for people with diabetes and at ages 46 to 47 years for those with metabolic syndrome, while aCR_{45} was reached at ages 41 to 43 for those with diabetes and at ages 41 and 42 for those with metabolic syndrome. Notably, the country-specific starting ages for those with diabetes or metabolic syndrome did not significantly deviate from the overall (sub)continental starting ages, with differences of no more than one year.

4 | Discussion

Due to the significant rise in early-onset CRC in Europe and North America, there is a need for tailored screening strategies to improve early detection and patient outcomes. Our analysis illustrates how disease-specific risk data can guide the

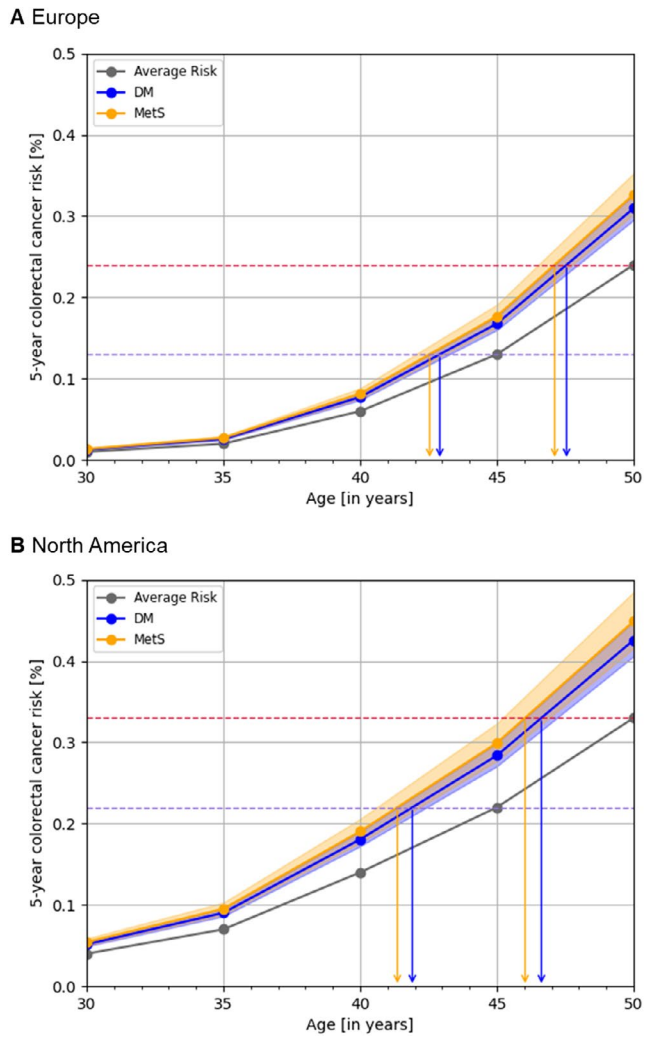


FIGURE 1 | Derivation of risk-adapted starting ages of colorectal cancer screening for people with diabetes and metabolic syndrome in Europe (A) and North America (B): Analyses based on relative risk estimates for CRC at any age with the comparison of average risk at 45 (purple line) and 50 years (red line). CRC, colorectal cancer; DM, diabetes; MetS, metabolic syndrome.

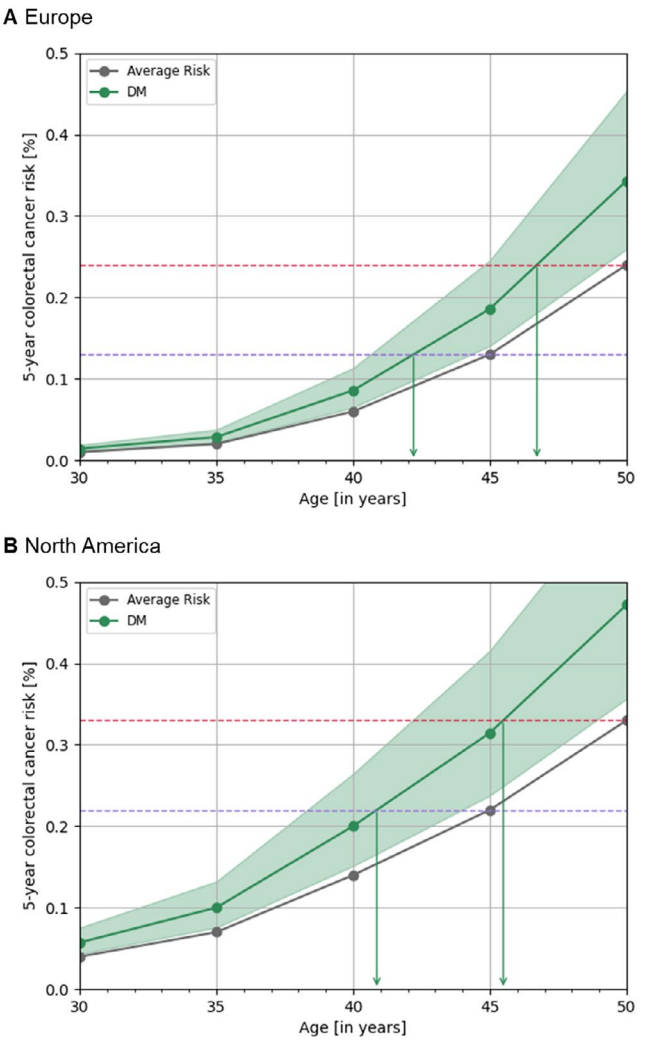


FIGURE 2 | Derivation of risk-adapted starting ages of colorectal cancer screening for people with diabetes in Europe (A) and North America (B): Analyses based on specific relative risk estimates for early-onset CRC with the comparison of average risk at 45 (purple line) and 50 years (red line). CRC, colorectal cancer; DM, diabetes.

determination of risk-adapted starting ages for CRC screening, potentially advancing earlier detection and improved outcomes. Specifically, we focused on the association of diabetes and metabolic syndrome with the increased risk of overall and early-onset CRC. Using the 5-year cumulative risk of the average risk population at age 50 as the benchmark, our analyses suggest that people with diabetes should initiate CRC screening three to four years earlier. The suggested advancement of the starting age remained consistent when considering the recently updated screening guidelines in the US, which recommends starting screening for the average-risk population at age 45. Similar to diabetes, people with metabolic syndrome would need to begin screening three to four years earlier to align with the benchmark screening age of 50, and two to four years earlier when considering the benchmark age of 45.

Two previous studies have derived risk-adapted starting ages for CRC screening for individuals with metabolic disorders. A nationwide cohort study based on registry data from Sweden estimated that those with diabetes reach the benchmark CRC risk at age 50 by the time they are 45, suggesting that their screening should start five years earlier than those at average risk [23]. A study based on data from the Cancer Screening Programme in Urban China [24] likewise reported that men and women with diabetes reach the same CRC risk level as those at average risk several years earlier. However, confidence intervals for these estimates were either not presented [23] or very broad [24]. Hence, our estimates which were derived from an up-to-date comprehensive international database on CRC incidence and the most comprehensive up-to-date risk estimates for diabetes and metabolic syndrome align with and strongly expand limited previous evidence by providing risk-adapted starting ages for large populations at high levels of precision.

Detection and removal of advanced neoplasms are critical in preventing CRC, highlighting the importance of assessing risk factors that increase the prevalence of these neoplasms [25]. Most CRC screening guidelines recommend that both men and women at average risk start screening at age 50, based on earlier

TABLE 3 | Risk-adapted starting ages for colorectal cancer screening.

	Risk-adapted starting age [years (95% confidence interval)] according to starting age in average-risk population, DM or MetS					
	Average risk: 50 years			Average risk: 45 years		
Country	DM[a]	DM[b]	MetS[a]	DM[a]	DM[b]	MetS[a]
Europe	48 (47–48)	47 (45–49)	47 (47–48)	43 (43–43)	42 (41–44)	43 (42–43)
Germany	48 (47–48)	47 (45–49)	47 (47–48)	43 (42–43)	42 (40–44)	42 (42–43)
France	47 (47–48)	46 (44–49)	47 (46–47)	42 (42–43)	42 (39–44)	42 (41–43)
United Kingdom	48 (47–48)	47 (45–49)	47 (47–48)	43 (42–43)	42 (40–44)	42 (42–43)
Italy	47 (47–48)	46 (44–49)	47 (46–48)	43 (42–43)	42 (40–44)	42 (42–43)
North America	47 (46–47)	46 (42–49)	46 (46–47)	42 (41–42)	41 (38–44)	41 (41–42)
United States	47 (46–47)	46 (43–49)	46 (46–47)	42 (41–42)	41 (38–44)	41 (41–42)
Canada	47 (46–47)	46 (42–49)	46 (45–47)	42 (42–43)	41 (39–44)	42 (41–42)

Abbreviations: DM, diabetes; MetS, metabolic syndrome.
[a]applying relative risk estimates for CRC at any age.
[b]applying relative risk estimates for early-onset CRC.

research showing a strong rise in CRC incidence and the emergence of advanced neoplasms from this age [26, 27]. However, recent shifts in epidemiological trends have prompted a re-evaluation of these guidelines. In response to these changes in CRC incidence, the US Preventive Services Task Force updated its guidelines, now recommending that screening should begin at 45 years for individuals at average risk [7]. This adjustment reflects a broader recognition that certain population subgroups may face elevated risks earlier than previously understood. Significant differences in early-onset CRC risk between North American and European populations require further exploration. Ethnic- and race-based disparities, or genetic variations in susceptibility to CRC or diabetes, may contribute to differing risk levels [28]. Furthermore, modifiable risk factors such as obesity, alcohol consumption and diabetes prevalence may influence early-onset CRC rates and show significant disparities, particularly between the US and Europe: Obesity is twice as prevalent in the US as in Europe [29]; alcohol consumption has significantly declined in Italy and France, while it has remained unchanged in Canada and the US [30, 31]; diabetes prevalence is higher in North America (11.9%) compared to Europe (7.0%) and is projected to rise to 24% and 9.2%, respectively, by 2045 [32]. Additionally, attitudes towards screening and differences in screening uptake and adherence could impact these disparities [28]. Understanding these factors is crucial for developing tailored screening guidelines and improving CRC prevention strategies across diverse populations and should therefore be the subject of future research [29].

Our analysis utilised the benchmark age of 45 and 50 years to accommodate for variations in starting ages as recommended by different CRC screening guidelines. The consistent patterns observed in our findings demonstrate the flexibility of our approach, allowing us to align with varying national screening guidelines, for which initial ages range from 45 years in the US to 50 in most European countries and Canada, or even 60 years in some parts of the UK [6, 33]. This adaptability demonstrates the potential to revise and harmonise international screening standards to accommodate varying risk levels across populations. Further, incorporating additional risk factors into CRC screening algorithms presents opportunities to enhance the precision of prevention strategies. Modifiable behaviours such as smoking, alcohol consumption, and a sedentary lifestyle are known to influence CRC risk [34]. By developing a comprehensive risk score that includes these lifestyle factors alongside biological indicators like age and familial history, it becomes possible to refine screening recommendations further [35]. Such an integrated approach could not only personalise prevention strategies more effectively but also could motivate individuals to take proactive steps in managing their health risks. For instance, recognising metabolic disorders as a significant risk factor in CRC screening guidelines may elevate its importance, encouraging individuals at risk for those diseases to adopt healthier lifestyle habits.

The findings of this study provide public health policymakers and clinicians with a basis to shift from predominantly fixed-age screening recommendations to a risk-adapted approach for high-risk individuals, facilitating earlier detection and potentially improving outcomes. While risk-adapted screening has been shown to be cost-effective when compared to no screening, inconsistencies in the literature make it challenging to definitively conclude whether risk-based CRC screening is more cost-effective and feasible than age-based screening [36]. Many of the evaluated approaches for risk-based screening rely on additional testing like genetic testing, which may increase logistical complexity and costs [36]. In contrast, approaches using routinely diagnosed clinical factors such as diabetes or metabolic syndrome can offer a more practical and resource-efficient alternative [8]. This approach could be compared to family history-based screening, which similarly focuses on a single routinely assessed factor and has been shown to be cost-effective compared to age-based screening [37, 38].

The primary strength of our study is the comprehensive data source utilised, providing a robust basis for our analysis.

Nonetheless, several limitations must be acknowledged to contextualise our findings accurately. First, our reliance on meta-analyses, despite their ability to synthesise data across multiple studies, may obscure heterogeneities in study design, population characteristics, and outcomes of the included studies. Although the meta-analyses attempted to control for other risk factors, confounding factors might still influence outcomes. Moreover, the definition of diabetes and metabolic syndrome varies, with different types of diabetes, country-specific metabolic syndrome criteria, and diverse treatment modalities (such as metformin or hypoglycaemic therapy) influencing risk reduction [39]. Additionally, the duration of diabetes is often underexplored, impacting the association with CRC risk. These factors introduce heterogeneity and potential biases in meta-analyses, complicating the synthesis and interpretation of findings. Furthermore, diabetes and metabolic syndrome share common underlying mechanisms, but their potential synergistic impact on CRC risk is not well understood and requires further research. Moreover, the lack of a dedicated meta-analysis exploring the relationship between metabolic syndrome and early-onset CRC limits our ability to draw definitive conclusions about this particular association. It appears plausible to assume that the relationship between metabolic syndrome and CRC risk may likewise be stronger at younger ages, which would result in even younger risk-adapted starting ages for screening. Future research should incorporate country- and age-specific data, once available, in more differentiated analyses regarding the implications of reducing the starting age for CRC screening below 45 or 50 years.

In conclusion, our analysis provides a straightforward approach for translating findings from comprehensive meta-analyses of CRC risk factors into a novel strategy for determining risk-adapted starting ages for CRC screening. Tailoring screening recommendations based on individual risk factors such as diabetes and metabolic syndrome could lead to more effective screening strategies for the increasing population at elevated risk of CRC, especially early-onset CRC. This approach may have the potential to improve detection rates, reduce the incidence of advanced disease, and optimise healthcare resource utilisation. Our analyses for diabetes and metabolic syndrome may serve as a model for determining risk-adapted starting ages for other risk factors and their combinations, e.g., as reflected in more comprehensive CRC risk scores. Further research, including comprehensive modelling studies, should evaluate the effectiveness and cost-effectiveness of such risk-adapted screening strategies.

Author Contributions

T.S. and H.B. conceptualised the study and designed the methodology. TS performed the data analysis and wrote the original draft of the manuscript; all authors contributed to its review and editing. HB supervised the study and acquired funding. All authors had access to the data and reviewed, verified, and consented to the published version of the manuscript. All accept the responsibility to submit the manuscript for publication.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data used for the analyses are publicly available from Global Cancer Observatory (GLOBOCAN): Cancer Today (version 1.1). Lyon, France: International Agency for Research on Cancer https://gco.iarc.fr/today/.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.